

current shift value by 1/256th of the clock period. The input signals to the PS unit are PSINCDEC, PSEN and PCLK. The output signals are PSDONE and the STATUS[0].

The SL unit indicates the current state of the DCM via the LOCKED and STATUS[0], STATUS[1] and STATUS[2] output signals. The LOCKED output signal indicates whether the

DCM outputs are in phase with the CLKIN input or not. The STATUS output signals indicate the state of the DLL and PS operations. The RST input signal resets the DCM logic and returns it to its post-configuration state.

Fig. 2 shows the DCM primitive.

The connection ports of the DCM are described in Table II.

TABLE II
Connection Ports of the DCM

Port	Direction	Description
CLKIN	Clock input	Clock input to DCM
CLKFB	Input	Clock feedback input to DCM
RST	Input	Asynchronous reset input 0 - No effect 1 - Reset DCM block
PSEN	Input	Dynamic phase shift enable 0 - Disable dynamic phase shifter 1 - Enable dynamic phase shifter
PSINCDEC	Input	Increment/decrement dynamic phase shift 0 - Decrement phase-shift value 1 - Increment phase-shift value
PCLK	Clock input	Clock input to dynamic phase shifter
CLK0	Clock output	Same frequency as CLKIN, 0-degree phase shift
CLK90	Clock output	Same frequency as CLKIN, 90-degree phase shift
CLK180	Clock output	Same frequency as CLKIN, 180-degree phase shift
CLK270	Clock output	Same frequency as CLKIN, 270-degree phase shift
CLK2X	Clock output	Double frequency clock output, 0-degree phase shift
CLK2X180	Clock output	Double frequency clock output, 180-degree phase shift
CLKDV	Clock output	Divided clock output
CLKFX	Clock output	Synthesised clock output
CLKFX180	Clock output	Synthesised clock output, 180-degree phase shift
STATUS[0]	Output	Dynamic phase-shift overflow 0 - phase shifter has not reached its limit value 1 - phase shifter has reached its limit value
STATUS[1]	Output	CLKIN input stopped indicator 0 - CLKIN input is toggling 1 - CLKIN input is not toggling
STATUS[2]	Output	CLKFX and CLKFX180 outputs stopped indicator 0 - CLKFX and CLKFX180 outputs are toggling 1 - CLKFX and CLKFX180 outputs are not toggling
STATUS[7:3]	Output	Reserved
LOCKED	Output	0 - DCM is attempting to lock onto CLKIN frequency and DCM clock outputs are not valid 1 - DCM is locked onto CLKIN frequency. DCM clock outputs are valid 1-to-0 - DCM lost lock, therefore reset DCM.
PSDONE	Output	Dynamic phase-shift operation complete 0 - No phase-shift operation active or phase-shift operation in progress 1 - Requested phase-shift operation is complete

Using DCM

Let us see how to use the DCM wizard to configure and instantiate the DCM module.

The design uses the DCM wizard to divide the input clock of 50MHz to 100kHz, which is then used to drive other circuits. One can use DCM for other applications mentioned above, including deskewing, clock phase shifting and so on.

Steps for using the DCM wizard

1. Create an integrated software environment (ISE) project.

(i) Launch an ISE. Select Start → Programs → Xilinx ISE Design Tools 13.4_1 → ISE Design Tools → Project Navigator

(ii) In Project Navigator, select File → New Project. The new project wizard opens.

(iii) In the location option, select the folder you want to work in. (We are working in the location C:/DCM. Select C:/DCM.) Click OK.

(iv) In Project name, type DCM_ClockDivider.

(v) Keep the top-level source type as HDL.

(vi) Click the Next button.

(vii) Select the family and the FPGA device type and other parameters in accordance with the FPGA and other tools you have. (We have used Virtex 5 series FPGA family, device type number XC5VLX50, package as FF676 and speed grade -1.) The project settings window looks as shown in Fig. 3.

(viii) Click Next and then Finish buttons.

2. In the Project tab, select New Source and the New Source Wizard will appear.

3. Select VHDL module and type ClockDivider in the File name field.

4. Check if Add to Project option is

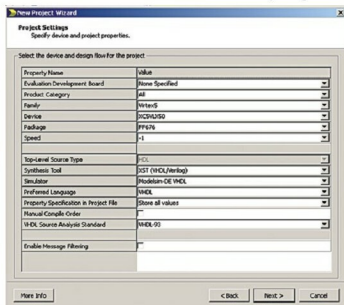


Fig. 3: Project Settings window